

Weekly Assessment – AI Generation & Validation System

1. Introduction

This document explains the implementation of an AI-based weekly assessment system. The system automatically generates questions, evaluates answers, and provides scoring without manual intervention.

2. Concept Overview

The system uses Artificial Intelligence to create weekly assessments based on the learning content (videos) consumed by trainees. It also validates answers using AI and provides instant results.

3. System Workflow

Step 1: Collect weekly learning data (videos/scripts).

Step 2: AI generates questions based on the content.

Step 3: Trainee attempts the assessment.

Step 4: System validates answers using AI (RAG).

Step 5: System calculates score and displays results.

4. Automated Question Generation

AI generates unique sets of questions every week based on the content learned. These include MCQs, True/False, and subjective questions.

5. Answer Validation (RAG)

The system uses Retrieval Augmented Generation (RAG) to validate answers. It compares user answers with stored learning material to ensure accuracy and relevance.

6. Scoring System

Objective Questions: Automatically scored based on correct answers.

Subjective Questions: AI evaluates answers based on meaning, logic, and understanding instead of just keywords.

7. Free vs Paid Implementation

Free Version:

- Hardcoded or basic AI-generated questions
- Simple keyword-based validation
- Basic scoring system
- Limited automation

Paid Version:

- Advanced AI-generated questions (OpenAI/GPT)
- Smart answer validation using RAG
- Semantic scoring for subjective answers
- Fully automated workflow with high accuracy

8. Technologies Used

- AI: OpenAI / Gemini
- Backend: Node.js
- Frontend: React
- Database: SQL / NoSQL
- Optional: Vector DB for advanced RAG

9. Key Benefits

- No manual effort required
- Instant evaluation and feedback
- Scalable for large number of users
- Personalized assessment

10. Conclusion

The AI-based weekly assessment system improves efficiency and accuracy by automating the entire evaluation process. It ensures consistent quality and provides a modern learning experience.